

R for bioinformatics, data summarisation and statistics

HUST Bioinformatics course series

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Section 1

section 1: TOC

前情提要

- basic plot functions
- basic ggplot2
- special letters
- equations
- advanced ggplot2

本次提要

- data summarisation functions (vector data)
 - median, mean, sd, quantile, summary
- 图形化的 data summarisation (two-D data/ tibble/ table)
 - dot plot
 - smooth
 - linear regression
 - correlation & variance explained
 - grouping & bar/ box/ plots
- statistics
 - parametric tests
 - t-test
 - one way ANNOVA
 - two way ANNOVA
 - linear regression
 - model / prediction / coefficients

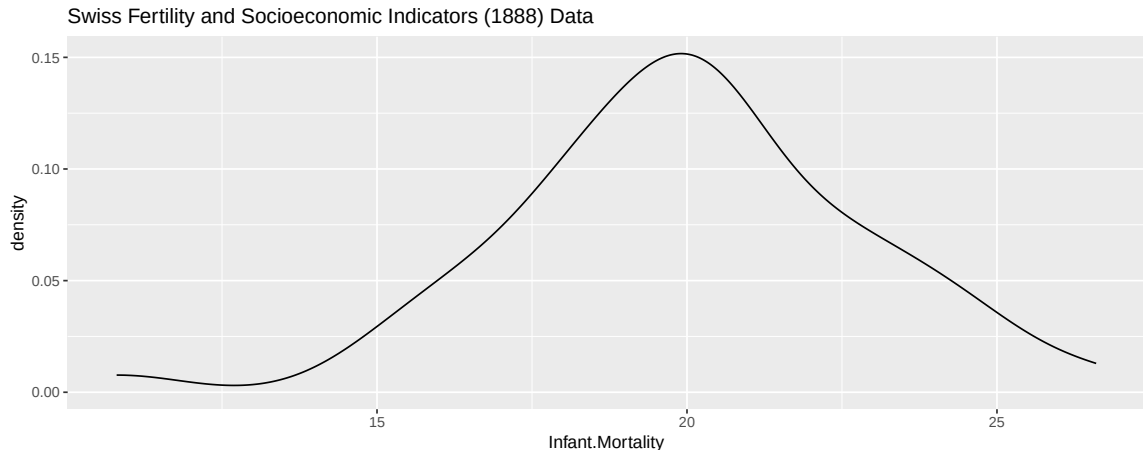
Section 2

section 2: vector summarisation

vector data

1 distribution

```
library(tidyverse);  
ggplot( swiss, aes( x = Infant.Mortality ) ) + geom_density() +  
  ggtitle("Swiss Fertility and Socioeconomic Indicators (1888) Data")
```



describe normal distributions

可以用 mean 和 sd 来描述

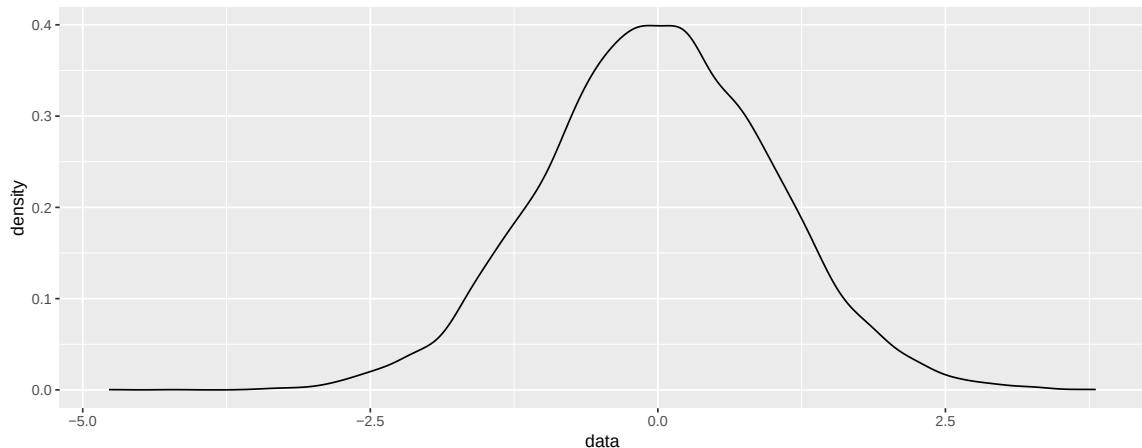
- Ⓐ It's symmetrical.
- Ⓑ Mean and median are the same.
- Ⓒ Most common values are near the mean; less common values are farther from it.
- Ⓓ Standard deviation marks the distance from the mean to the inflection point.

$(\text{mean} + 1 * \text{sd}) \geq 68\%$

$(\text{mean} + 2 * \text{sd}) \geq 95\%$ 的数据

functions to generate random normal distributions

```
# 生成 10000 个随机数字, 使其 mean = 0, sd = 1, 且为 normal distribution ...  
x <- rnorm(10000, mean = 0, sd = 1);  
ggplot( data.frame( data = x ), aes( data ) ) + geom_density( );
```



More to read: http://uc.r.githuh.io/generating_random_numbers/

other functions to generate random normal distributions

注意, 以下函数中的 q , p , x 需要自行提供

```
# generate CDF probabilities for value(s) in vector q
pnorm(q, mean = 0, sd = 1)

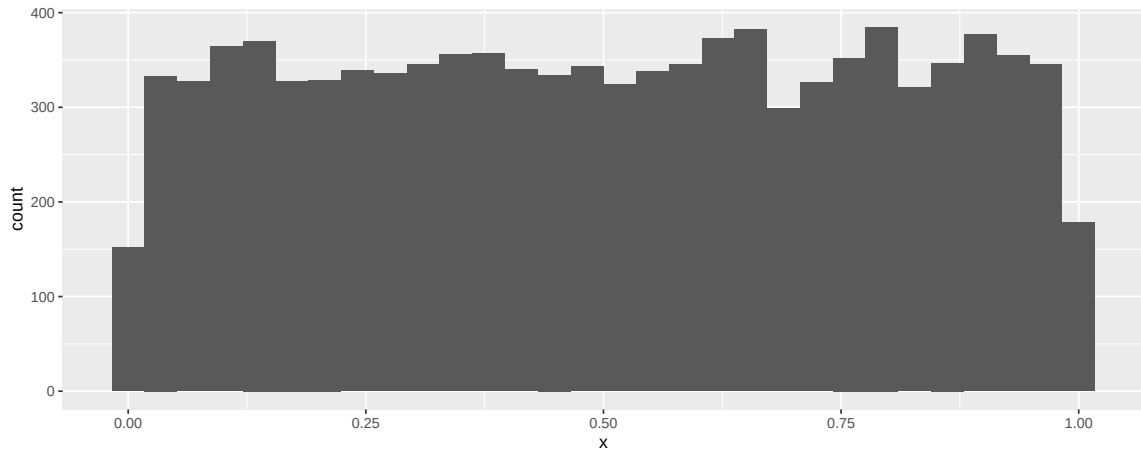
# generate quantile for probabilities in vector p
qnorm(p, mean = 0, sd = 1)

# generate density function probabilities for value(s) in vector x
dnorm(x, mean = 0, sd = 1)
```

其它规律的 distributions

① uniform distributions

```
x <- runif( 10000 ); ## random numbers of uniform distributions between 0 and 1  
ggplot( data.frame( dat = x ), aes( x ) ) + geom_histogram();
```



uniform distribution 的各种函数

注: 以下函数中的 n 需要自行决定

```
# generate n random numbers between 0 and 25
runif(n, min = 0, max = 25)

# generate n random numbers between 0 and 25 (with replacement)
sample(0:25, n, replace = TRUE)

# generate n random numbers between 0 and 25 (without replacement)
sample(0:25, n, replace = FALSE)
```

other distributions, cont.

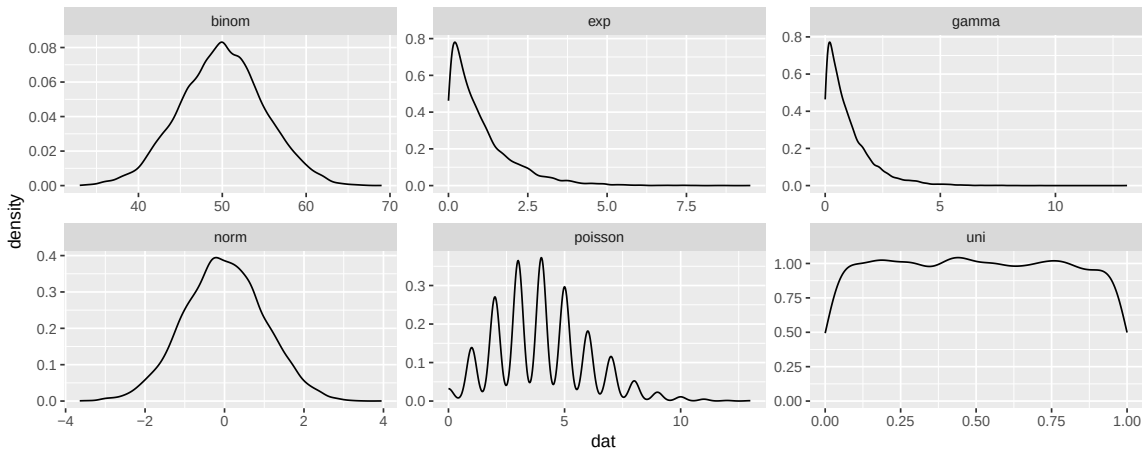
```
n <- 10000;
uni <- tibble( dat = runif(n), type = "uni" );
norm <- tibble( dat = rnorm(n), type = "norm" );
binom <- tibble( dat = rbinom(n, size = 100, prob = 0.5), type = "binom" );
poisson <- tibble( dat = rpois(n, lambda = 4), type = "poisson" );
exp <- tibble( dat = rexp(n, rate = 1) , type = "exp");
gamma <- tibble( dat = rgamma(n, shape = 1) , type = "gamma");

combined <- bind_rows( uni, norm, binom, poisson, exp, gamma );

plot1 <-
  ggplot( combined , aes( dat ) ) + geom_density() +
    facet_wrap( ~type, ncol = 3, scales = "free");
```

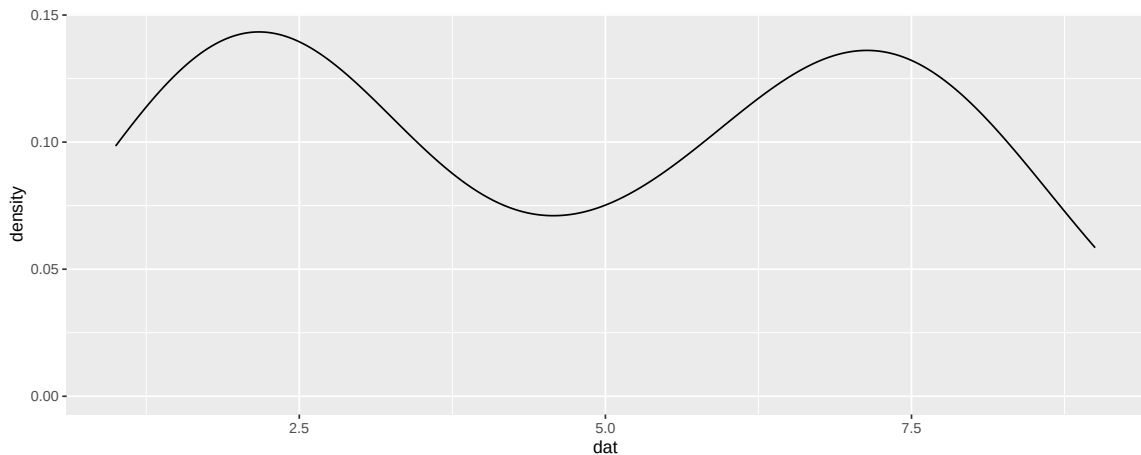
other distributions, plot

```
plot1;
```



non-parametric distribution

```
## votes on people's desire to visit  
bi <- c(7, 3, 2, 1, 7, 3, 4, 5, 7, 6, 2, 2, 1, 3, 7, 2, 6, 8, 2, 7, 2, 2, 1,  
3, 5, 8, 2, 6, 7, 8, 6, 2, 8, 7, 9, 2, 7, 5, 1, 8, 8, 2, 3, 7, 3, 8);  
ggplot( data.frame( dat = bi ), aes(dat)) + geom_density();
```



量化描述数据

使用以下同名函数

mean: aka average, is the sum of all of the numbers in the data set divided by the size of the data set.

median: The median is the value that is in the middle when the numbers in a data set are sorted in increasing order.

sd: standard deviation

var: measures how far a set of numbers are spread out

range: 取值范围

note: from:

<https://www.ai-therapy.com/psychology-statistics/descriptive/mean-mode-median>



量化描述函数

```
mean( norm$dat );  
median( norm$dat );  
## mode( norm$dat ); ## ???  
  
sd(norm$dat);  
var(norm$dat);  
range(norm$dat);
```

quantile and summary

```
quantile( norm$dat );
```

0%	25%	50%	75%	100%
-3.64368738	-0.68690036	-0.01630631	0.66562044	3.95352376

quantile 还接受其它参数

```
quantile( norm$dat, probs = seq(0, 1, length = 11));
```

0%	10%	20%	30%	40%	50%
-3.64368738	-1.28394614	-0.86116054	-0.53025196	-0.27164280	-0.01630631
60%	70%	80%	90%	100%	
0.24608754	0.52618066	0.82947721	1.29876407	3.95352376	

summary ...

```
summary( norm$dat );
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-3.643687	-0.686900	-0.016306	-0.007199	0.665620	3.953524

summary 也可应用于非数值

```
summary( combined$type );
```

Length	Class	Mode
--------	-------	------

summary, cont.

```
## summary 可应用于整个表格；相当于对每列进行 summary ...
summary( combined );
```

dat	type
Min. : -3.6437	Length: 60000
1st Qu.: 0.3128	Class : character
Median : 0.9403	Mode : character
Mean : 9.4133	
3rd Qu.: 4.0000	
Max. : 69.0000	

table 函数

返回 vector 当中 unique 值和它们的出现次数

```
table( combined$type );
```

```
binom      exp      gamma      norm poisson      uni
10000    10000    10000    10000    10000    10000
```

**** 注 ****: table 还接受 data.frame 作为输入, 比如 table(combined)。请自行尝试并理解结果

count in dplyr

```
combined %>% dplyr::count( type );
```

```
# A tibble: 6 x 2
  type      n
  <chr>  <int>
1 binom  10000
2 exp    10000
3 gamma  10000
4 norm   10000
5 poisson 10000
6 uni    10000
```

which is similar to:

```
combined %>% group_by(type) %>% count( name = "my_count" );
```

```
# A tibble: 6 x 2
# Groups:   type [6]
  type    my_count
  <chr>    <int>
1 binom    10000
2 exp      10000
3 gamma    10000
```

dplyr::count cont.

```
# For table()-like output with two factors:
```

```
iris %>%
  group_by(Species) %>%
  count(Petal.Width)
```

```
# A tibble: 27 x 3
```

```
# Groups:   Species [3]
```

	Species	Petal.Width	n
	<fct>	<dbl>	<int>
1	setosa	0.1	5
2	setosa	0.2	29
3	setosa	0.3	7
4	setosa	0.4	7
5	setosa	0.5	1
6	setosa	0.6	1
7	versicolor	1	7
8	versicolor	1.1	3
9	versicolor	1.2	5
10	versicolor	1.3	13

```
# i 17 more rows
```

Section 3

section 3: two column data: part 1

数据介绍: a numeric vector and a factorial vector

此类数据，通常一列是数值，另一列是分组信息，如下例：

```
data.fig3a <- read_csv( file = "data/talk10/nc2015_data_for_fig3a.csv" );
```

```
head( data.fig3a[ c("tai", "trans.at") ] ); ## 只显示有用的两列
```

```
# A tibble: 6 x 2
```

	tai	trans.at
	<dbl>	<dbl>
1	1	0.195
2	1	-0.320
3	1	-0.327
4	1	-0.304
5	1	-0.275
6	1	-0.167

数据介绍, cont.

tai: 表达量的一种计算方式, 1 == lowest, 5 == highest

trans.at: A - T 碱基使用偏好;

假说:

- (a) de novo synthesis cost of A is higher than T,
- (b) therefore highly expressed genes will tend to use T than A when possible.
- (c) 因此, 在高表达的基因当中, A - T 的差值会变大 (更负)。

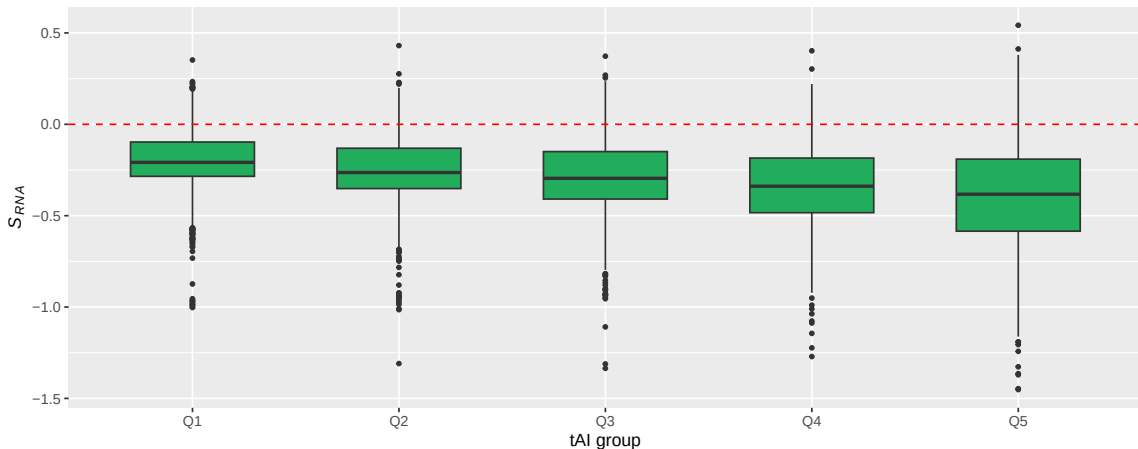
data source: Chen et al, Nature Communications, 2016

boxplot

```
fig3a <-
  ggplot( data.fig3a, aes( factor(tai), trans.at ) ) +
    geom_boxplot( fill = "#22AD5C", linetype = 1 ,outlier.size = 1, width = 0.6) +
    xlab( "tAI group" ) +
    ylab( expression( paste( italic(S[RNA]) ) ) ) +
    scale_x_discrete(breaks= 1:5 , labels= paste("Q", 1:5, sep = "")) +
    geom_hline( yintercept = 0, colour = "red", linetype = 2);
```

show the plot

```
fig3a;
```



说明:

- ① 此种情况下，我们通常只看 median 值的趋势;

another example plot

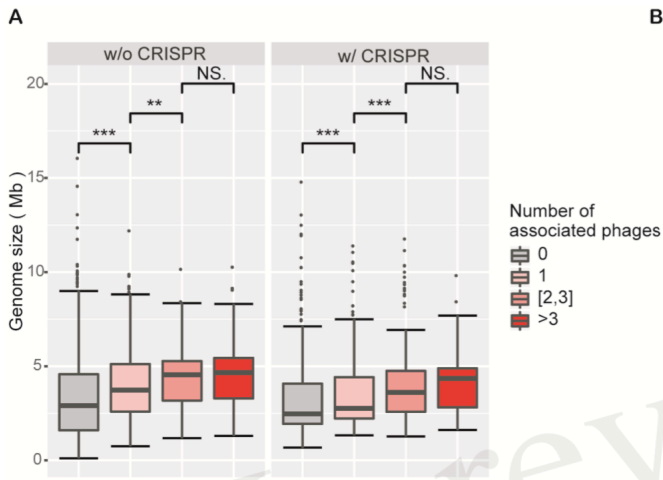
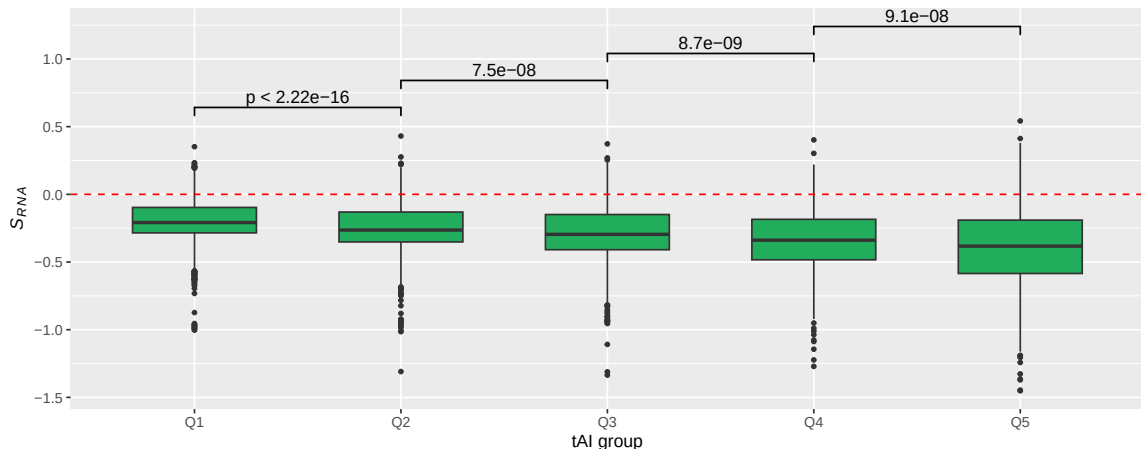


Figure 1: a figure from an in-press manuscript

how to add significance indicators to plot??

ggplot2 的扩展包, `geom_signif`; 如果第一次使用, 请先安装。

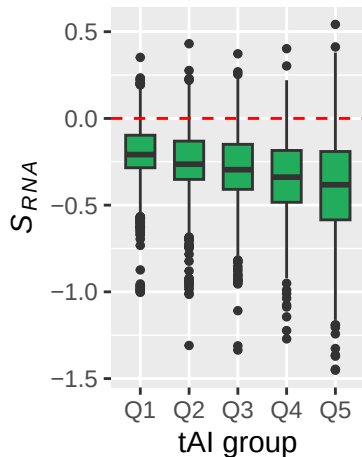
```
library(ggsignif);
fig3a + geom_signif( comparisons = list(1:2, 2:3, 3:4, 4:5), test = wilcox.test,
                      step_increase = 0.1 );
```



boxplot 画图注意事项

正确的画法为：要高瘦，不要矮胖！！！！

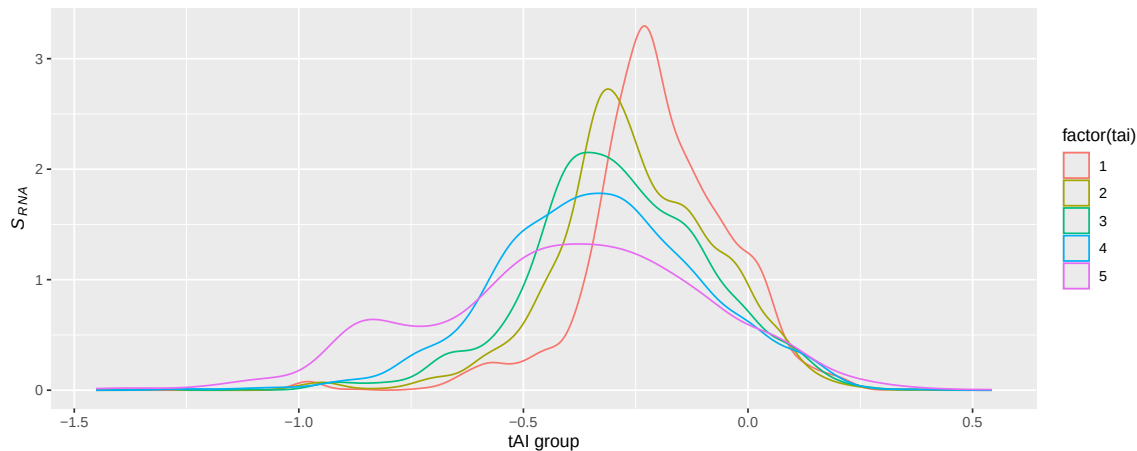
```
fig3a;
```



此类数据的另一种可视化方式

density plot; 但在此例中, 不如 boxplot 好。

```
ggplot( data.fig3a, aes( trans.at, colour = factor(tai) ) ) + geom_density( ) +  
  xlab( "tAI group" ) + ylab( expression( paste( italic(S[RNA]) ) ) );
```



Section 4

section 4: two column data: part 2

数据介绍: two numerical vectors

多用于描述两组（量化）数据之间的关系；

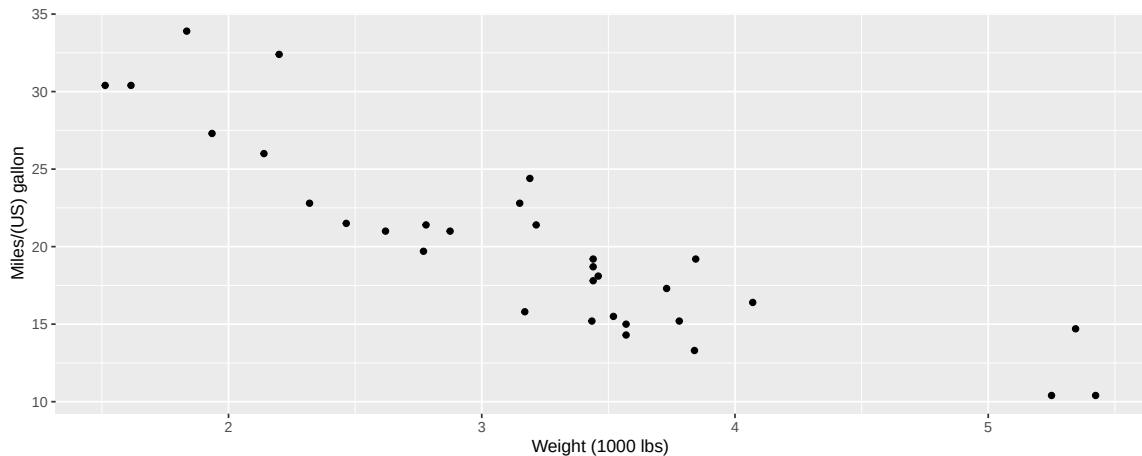
以 mtcars 为例:

```
head(mtcars);
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

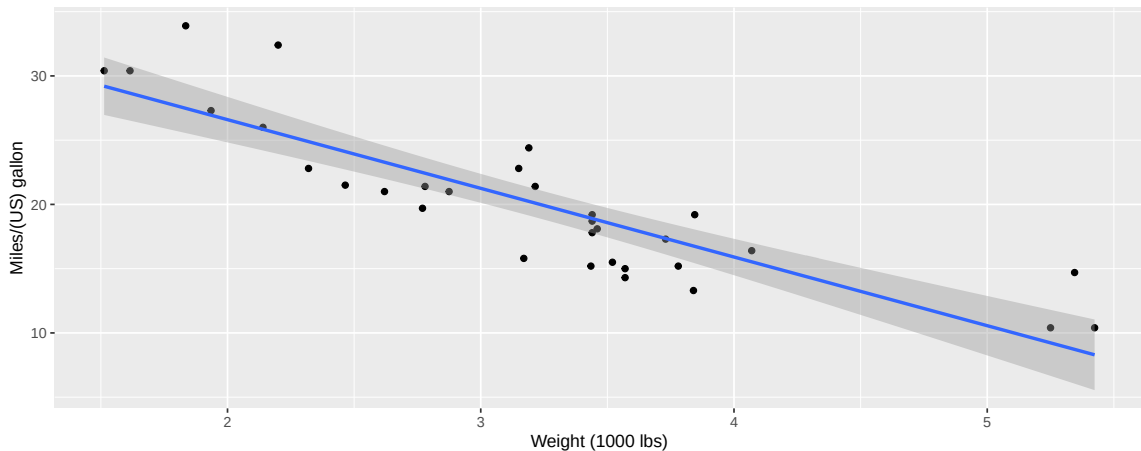
查看重量与燃油效率之间的关系

```
plotcars <-  
  ggplot( mtcars, aes( x = wt, y = mpg ) ) +  
    geom_point() + xlab( "Weight (1000 lbs)" ) + ylab( "Miles/(US) gallon" );  
plotcars;
```



smooth, 减少噪音

```
plotcars + geom_smooth( method = "lm" ); ## default is lowess
```



expression(R^2) variance of x explained by y ...

```
( r <- with( mtcars, cor.test( mpg, wt )$estimate ) );
```

```
      cor
-0.8676594
```

```
## variance of mpg can be explained by weight
r^2;
```

```
      cor
0.7528328
```

当趋势不明显时，可以按另一组数据分组

这里还以 mtcars 为例。

两种分组（binning）方法 equal-distance, equal-size binning

举例:

```
mtcars2 <- mtcars %>%
  mutate( group1 = ntile( wt, 4 ), ## equal-size binning
           group2 = cut( wt,
                         breaks = seq( from = min(wt), to = max(wt),
                                       by = (max(wt) - min(wt)) / 4 ),
                         include.lowest = T ) ## equal-distance ...
  ) ;
```

ntile 函数的参数

... *ntile 函数都是 equal size

```
## ntile 的结果
table( mtcars2$group1 );
```

```
1 2 3 4
8 8 8 8
```

cut 函数

按指定的间隔 (breaks) 对数据进行分割。

```
table( mtcars2$group2);
```

```
[1.51,2.49] (2.49,3.47] (3.47,4.45] (4.45,5.42]
           8          13           8           3
```

使用方法:

```
cut(x, ...)
# S3 method for default
cut(x, breaks, labels = NULL,
    include.lowest = FALSE, right = TRUE, dig.lab = 3,
    ordered_result = FALSE, ...)
```

cut 示例

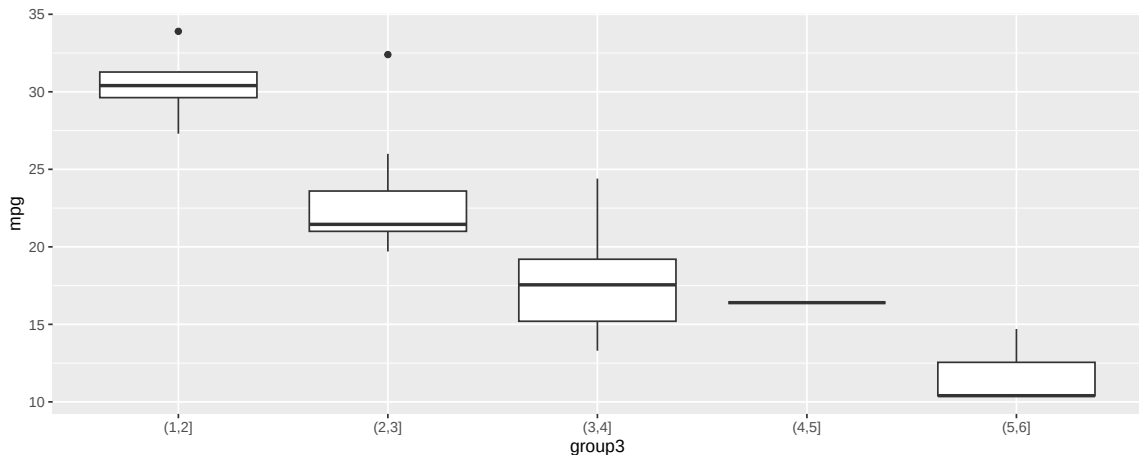
不仅可用于 equal distance, 还可以用于任意间距

```
mtcars3 <- mtcars2 %>%
  mutate( group3 = cut( mtcars$wt, breaks = c(0,1,2,3,4,5,6) ) ) ;
table(mtcars3$group3);
```

```
(0,1] (1,2] (2,3] (3,4] (4,5] (5,6]
    0     4     8    16     1     3
```


分组后的数据适合用 **boxplot**

```
ggplot( mtcars3, aes( group3, mpg ) ) +  
  geom_boxplot();
```



小结

目前讲述了以下内容:

一维数据

table, summary, range, quantile, mean, median ...

二维数据

- boxplot
- point plot
- correlation
- 分组: equal distance, equal size binning ...

Section 5

section 5: parametric tests

parametric tests

- ① 包括:
 - t-test
 - analysis of variance
 - linear regression
- ② 数据有较明确的分布 (e.g. normal distribution), 或假设数据有明确的分布; 当假设不成立时, 检测会无效;
- ③ 更灵敏 (相比 nonparametric test), p-value 更低

more to read: http://rcompanion.org/handbook/I_01.html

适用性

适用于

- 数量化性状, 比如: 身高、体重、产量、污染值
- 整数值: 成绩、年龄、每天步数

不适用于

- 其它 count data 或者 discrete data;
- 或者有太多趋向于 min 或 max 的值
- 百分比或比例

详见: <http://rcompanion.org/handbook/index.html>

需要的 packages

需要的 packages

```
## chooseCRANmirror()
if(!require(psych)) {
  install.packages("psych");
}
if( !require(rcompanion) ) {
  install.packages("rcompanion");
}

library(psych);
library(rcompanion)
```

数据

注意 source() 函数的用法

```
source("data/talk10/input_data1.R"); ## 装入 Data data.frame ...

str(Data);
```

```
'data.frame':  26 obs. of  5 variables:
 $ Student: chr  "a" "b" "c" "d" ...
 $ Sex    : chr  "female" "female" "female" "female" ...
 $ Teacher: chr  "Catbus" "Catbus" "Catbus" "Catbus" ...
 $ Steps  : int   8000 9000 10000 7000 6000 8000 7000 5000 9000 7000 ...
 $ Rating : int    7 10 9 5 4 8 6 5 10 8 ...
```

检查数据

```
library(psych)
headTail(Data); ## psych 包提供的函数
```

	Student	Sex	Teacher	Steps	Rating
1	a	female	Catbus	8000	7
2	b	female	Catbus	9000	10
3	c	female	Catbus	10000	9
4	d	female	Catbus	7000	5
...	<NA>	<NA>	<NA>	...	...
23	w	male	Totoro	6000	8
24	x	male	Totoro	8000	10
25	y	male	Totoro	7000	7
26	z	male	Totoro	7000	7

查看数据, cont.

```
## 其它常用函数
```

```
str(Data)
```

```
'data.frame':  26 obs. of  5 variables:
 $ Student: chr  "a" "b" "c" "d" ...
 $ Sex    : chr  "female" "female" "female" "female" ...
 $ Teacher: chr  "Catbus" "Catbus" "Catbus" "Catbus" ...
 $ Steps  : int  8000 9000 10000 7000 6000 8000 7000 5000 9000 7000 ...
 $ Rating : int   7 10  9  5  4  8  6  5 10  8 ...
```

```
summary(Data)
```

Student	Sex	Teacher	Steps
Length:26	Length:26	Length:26	Min. : 5000
Class :character	Class :character	Class :character	1st Qu.: 7000
Mode :character	Mode :character	Mode :character	Median : 8000
			Mean : 7692
			3rd Qu.: 8750
			Max. :10000
Rating			
Min. : 4.000			
1st Qu.: 7.000			

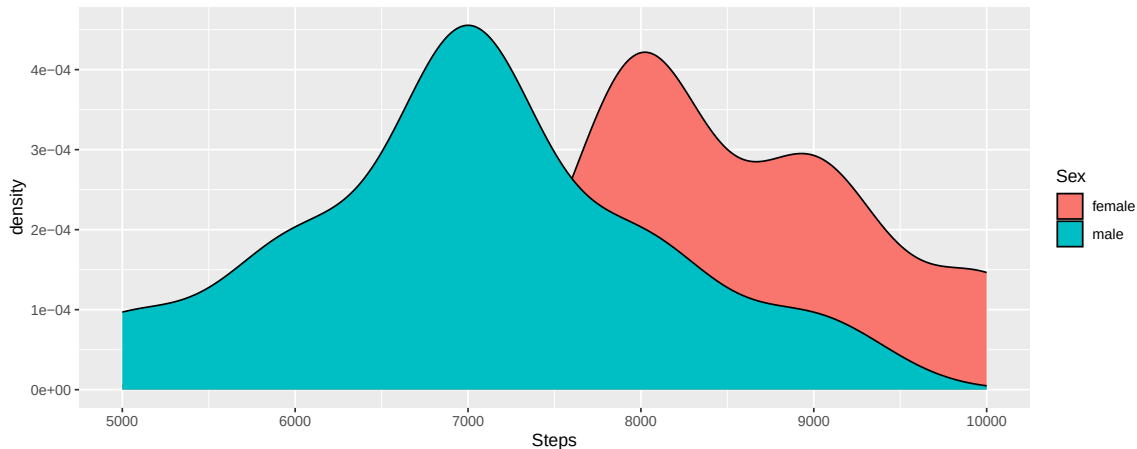


parametric test 的要求

- ① 随机取样
- ② 值或 residuals 为正态分布; residues 是指观察值与预测值 (mean) 之差

数据的分布

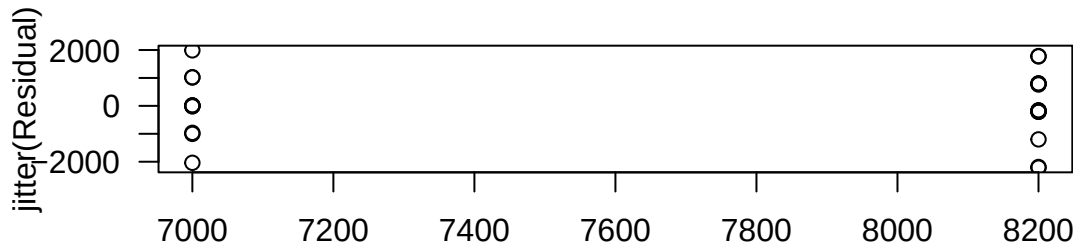
```
ggplot(Data, aes(Steps, fill = Sex)) +  
  geom_density(position="dodge", alpha = 1)
```



parametric test 的要求, cont.

③ 有相同的 variance

```
M1 = mean(Data$Steps[Data$Sex=="female"])
M2 = mean(Data$Steps[Data$Sex=="male"])
Data$Mean[Data$Sex=="female"] = M1
Data$Mean[Data$Sex=="male"] = M2
Data$Residual = Data$Steps - Data$Mean
plot(jitter(Residual) ~ Mean, data = Data, las = 1);
```



how to detect outlier ??

一个很模糊的定义: Outliers are extreme values that fall a long way outside of the other observations. For example, in a normal distribution, outliers may be values on the tails of the distribution.

对于 normal distribution, 通常 $\text{mean} \pm 2 \text{ or } 3 * \text{sd}$

对于 non-parametric distribution (注: IQR 计算可使用同名函数: IQR):

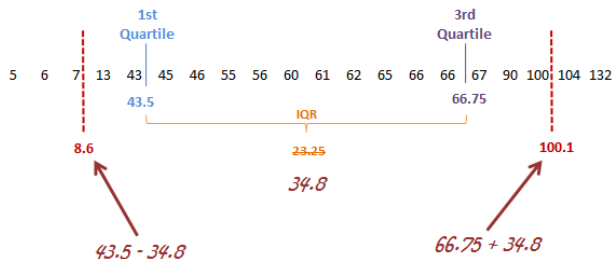
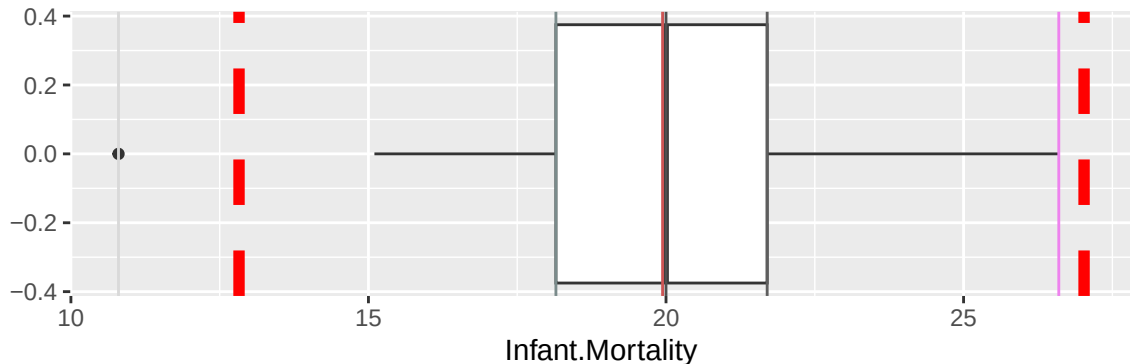


Figure 2: Tukey's method for outlier detection

an example of outlier values

```
s <- summary( swiss$Infant.Mortality );
iqr <- IQR(swiss$Infant.Mortality);
ggplot( swiss, aes( y = Infant.Mortality ) ) + geom_boxplot() + coord_flip() +
  geom_hline( yintercept = s, colour = sample( colors(), length(s) ) ) +
  geom_hline( yintercept = c( s["1st Qu."] - 1.5 * iqr, s["3rd Qu."] + 1.5 * iqr ),
    colour = "red", size = 2, linetype = 2);
```



one sample t-test

检测分布是否与预期一致; 比如: 男生每天的步数是否显著区别于 1 万

```
with( Data, t.test( Steps[ Sex == "male" ], mu = 10000 ) );
```

One Sample t-test

```
data:  Steps[Sex == "male"]
t = -9.083, df = 10, p-value = 3.81e-06
alternative hypothesis: true mean is not equal to 10000
95 percent confidence interval:
 6264.07 7735.93
sample estimates:
mean of x
 7000
```

two samples t-test

比较 sd 和 mean , 可应用于正态分布。几种使用方法:

```
with( Data, t.test( Steps ~ Sex ) )
```

Welch Two Sample t-test

```
data: Steps by Sex
```

```
t = 2.6424, df = 22.816, p-value = 0.01461
```

```
alternative hypothesis: true difference in means between group female and group male is not eq
```

```
95 percent confidence interval:
```

```
260.1421 2139.8579
```

```
sample estimates:
```

```
mean in group female    mean in group male
```

```
8200
```

```
7000
```


two sample t-test 使用方法 2

```
with( Data, t.test( Steps[ Sex == "male" ], Steps[ Sex == "female" ] ) );
```

Welch Two Sample t-test

data: Steps[Sex == "male"] and Steps[Sex == "female"]

t = -2.6424, df = 22.816, p-value = 0.01461

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-2139.8579 -260.1421

sample estimates:

mean of x mean of y

7000 8200

two sample t test 检测结果

```
res <- with( Data, t.test( Steps ~ Sex ) );
str(res);
```

List of 10

```
$ statistic : Named num 2.64
..- attr(*, "names")= chr "t"
$ parameter : Named num 22.8
..- attr(*, "names")= chr "df"
$ p.value : num 0.0146
$ conf.int : num [1:2] 260 2140
..- attr(*, "conf.level")= num 0.95
$ estimate : Named num [1:2] 8200 7000
..- attr(*, "names")= chr [1:2] "mean in group female" "mean in group male"
$ null.value : Named num 0
..- attr(*, "names")= chr "difference in means between group female and group male"
$ stderr : num 454
$ alternative: chr "two.sided"
$ method : chr "Welch Two Sample t-test"
$ data.name : chr "Steps by Sex"
- attr(*, "class")= chr "htest"
```



paired two sample t test

例如: 辅导前后的学生成绩:

```
source("data/talk10/input_data2.R");  
  
head(scores);
```

	Time	Student	Score
1	Before	a	65
2	Before	b	75
3	Before	c	86
4	Before	d	69
5	Before	e	60
6	Before	f	81

paired two sample t test

```
scores.wide <- scores %>% spread( Time, Score );
head(scores.wide, n = 3);
```

	Student	After	Before
1	a	77	65
2	b	98	75
3	c	92	86

```
with( scores.wide, t.test( After, Before, paired = T ) );
```

Paired t-test

data: After and Before

t = 3.8084, df = 9, p-value = 0.004163

alternative hypothesis: true mean difference is not equal to 0

95 percent confidence interval:

4.141247 16.258753

sample estimates:

mean difference

10.2

one way ANOVA

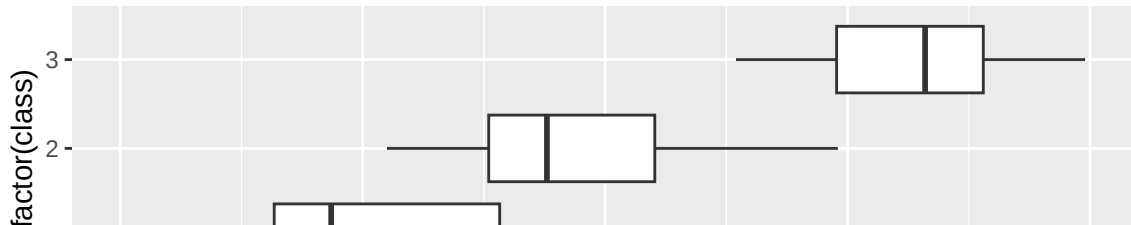
ANOVA: similar to independent t-test, but can be applied to multiple groups

主要分析一个变量（多组）对另一变量的贡献度/解释度/影响度。

比如：年级对学生的体重是否有影响？

准备：3 个班学生的体重

```
wt<- bind_rows( tibble( class = 1, wt = sample( seq(50, 60, by = 0.1), 20 ) ),
                 tibble( class = 2, wt = sample( seq(55, 65, by = 0.1), 20 ) ),
                 tibble( class = 3, wt = sample( seq(60, 70, by = 0.1), 20 ) )
               );
ggplot(wt, aes( factor( class ), wt ) ) + geom_boxplot() + coord_flip();
```



one way ANOVA, cont.

```
library(FSA); ## 如果没有这个包, 请先安装 ...
with( wts, Summarize( wt ~ class, digits = 3 ) );
```

	class	n	mean	sd	min	Q1	median	Q3	max
1	1	20	55.175	3.087	50.0	53.175	54.35	57.825	59.9
2	2	20	59.195	2.579	55.5	57.600	58.80	61.025	64.8
3	3	20	66.405	2.062	62.7	64.775	66.60	67.800	69.9

linear model

两个问题:

① 组间有显著区别吗?

```
model <- lm( wt ~ class, data = wts );
anova( model );
```

Analysis of Variance Table

Response: wt

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
class	1	1261.13	1261.13	173.27	< 2.2e-16 ***
Residuals	58	422.14	7.28		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



ANOVA

② 分组对变量的贡献 (r-square, aka. variance explained)

```
summary( model );
```

Call:

```
lm(formula = wt ~ class, data = wts)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.7583	-2.0083	-0.2933	2.1454	5.2567

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	49.0283	0.9215	53.21	<2e-16 ***
class	5.6150	0.4266	13.16	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.698 on 58 degrees of freedom

Multiple R-squared: 0.7492, Adjusted R-squared: 0.7449

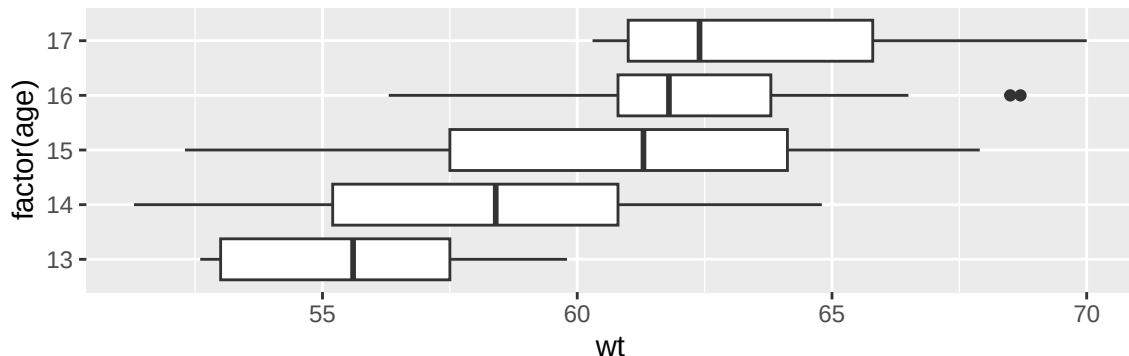
F-statistic: 173.2 on 1 and 58 DF, p-value: 6.28e-16



one way ANOVA with blocks

同时有多个因素影响体重时，哪些才是主要的？

```
wt2 <- bind_rows(
  tibble( class = 1, age = sample( 13:15, 20, replace = T ), wt = sample( seq(50, 60, by = 0.
  tibble( class = 2, age = sample( 14:16, 20, replace = T ), wt = sample( seq(55, 65, by = 0.
  tibble( class = 3, age = sample( 15:17, 20, replace = T ), wt = sample( seq(60, 70, by = 0.
);
ggplot(wt2, aes( factor( age ), wt ) ) + geom_boxplot() + coord_flip();
```



one way ANOVA with blocks, cont.

```
model2 <- lm( wt ~ class + age, data = wts2);
anova( model2 );
```

Analysis of Variance Table

Response: wt

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
class	1	758.64	758.64	83.2853	9.616e-13 ***
age	1	0.46	0.46	0.0507	0.8227
Residuals	57	519.21	9.11		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

如何获得 r.squre value ???

one way ANOVA with blocks, 各个 factor 的重要性??

```
library(relaimpo);
```

```
res3 <- calc.relimp( wt ~ factor(class) + age, data = wts2 );
res3$R2; ## 总 R2
```

```
[1] 0.5980402
```

```
res3$lmgi; ## 每个因素的贡献;
```

```
factor(class)          age
      0.4454314      0.1526088
```

```
## 测试 rela 参数:
```

```
res4 <- calc.relimp( wt ~ factor(class) + age, data = wts2, rela = T);
res4$R2; ## 总 R2
```

```
[1] 0.5980402
```

```
res4$lmgi; ## 每个因素的贡献;
```

```
factor(class)          age
      0.7448184      0.2551816
```

更多信息: http://rcompanion.org/handbook/L_06.html



two way ANOVA

一个变量受另外两个因素影响的分析; 比如上例中 体重 受 年级 和 年龄 的影响。

年级 和 年龄 至少有 4 个 unique combinations .

实际上, 上面的 block test 可以认为是 two-way ANOVA 分析

```
Summarize(wt ~ age + class, data = wts2, digits=3);
```

	age	class	n	mean	sd	min	Q1	median	Q3	max
1	13	1	7	55.571	2.871	52.6	53.000	55.60	57.50	59.8
2	14	1	6	54.350	2.272	51.3	52.875	54.30	55.95	57.3
3	15	1	7	56.986	2.807	52.3	55.500	57.00	59.35	59.9
4	14	2	11	60.509	3.190	55.2	59.250	60.30	62.75	64.8
5	15	2	3	63.467	0.862	62.7	63.000	63.30	63.85	64.4
6	16	2	6	59.717	3.077	56.3	57.000	59.95	61.70	63.8
7	15	3	4	65.150	2.002	63.1	64.300	64.80	65.65	67.9
8	16	3	7	64.614	3.300	60.8	62.050	63.70	67.50	68.7
9	17	3	9	63.922	3.657	60.3	61.000	62.40	65.80	70.0

two way ANOVA, cont.

```
model3 <- lm( wt ~ class + age + class:age, data = wts2);
anova( model3 );
```

Analysis of Variance Table

Response: wt

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
class	1	758.64	758.64	84.5867	8.674e-13 ***
age	1	0.46	0.46	0.0515	0.8213
class:age	1	16.96	16.96	1.8907	0.1746
Residuals	56	502.25	8.97		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

其中: class 和 age 称为 **main effects**, class:age 称为 **interaction effects**

relative importance of interactions

```
res5 <- calc.relimp( wt ~ factor(class) + age + factor(class):age, data = wts2);
res5$R2; ## 总 R2
```

```
[1] 0.6077766
```

```
res5$lm; ## 每个因素的贡献;
```

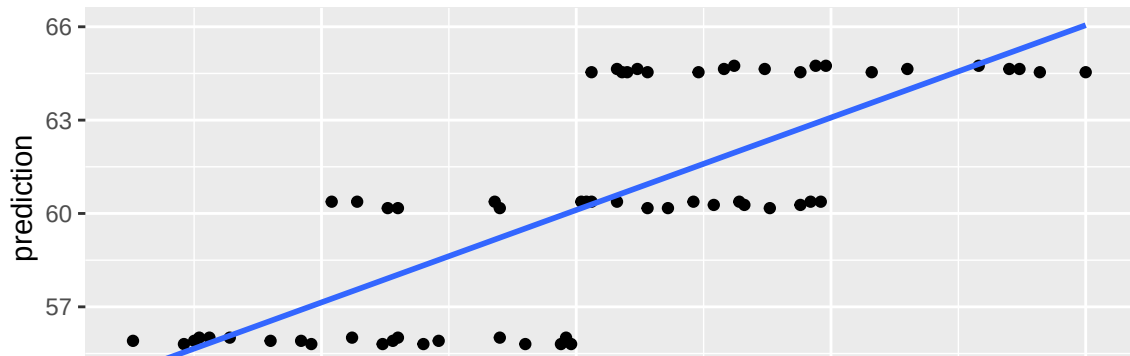
factor(class)	factor(class):age	age
0.44543136	0.00973635	0.15260884

用模型进行预测 predict

```
model2 = lm(formula = wt ~ class + age, data = wts2)

newdata <- wts2 %>% dplyr::select( class, age );
wt.predicted <- predict( model2, newdata );

dat <- data.frame( reference = wts2$wt, prediction = wt.predicted );
ggplot( dat , aes( x = reference, y = prediction ) ) + geom_point() +
  geom_smooth( method = "lm", se = F );
```



prediction 与 original data 的 correlation 是多少??

```
with( dat, cor.test( prediction, reference ) )$estimate;
```

```
cor
0.7706049
```

```
##  $R^2$ 
```

```
with( dat, cor.test( prediction, reference ) )$estimate ^2;
```

```
cor
0.5938319
```

```
## 正好是 model2 的 r.squared ...
```

```
summary( model2 )$r.squared;
```

```
[1] 0.5938319
```


手动计算 prediction

在一个 linear model 中, $wt = \text{intercept} + a * \text{class} + b * \text{age}$

而 intercept , a, b 的值分别为:

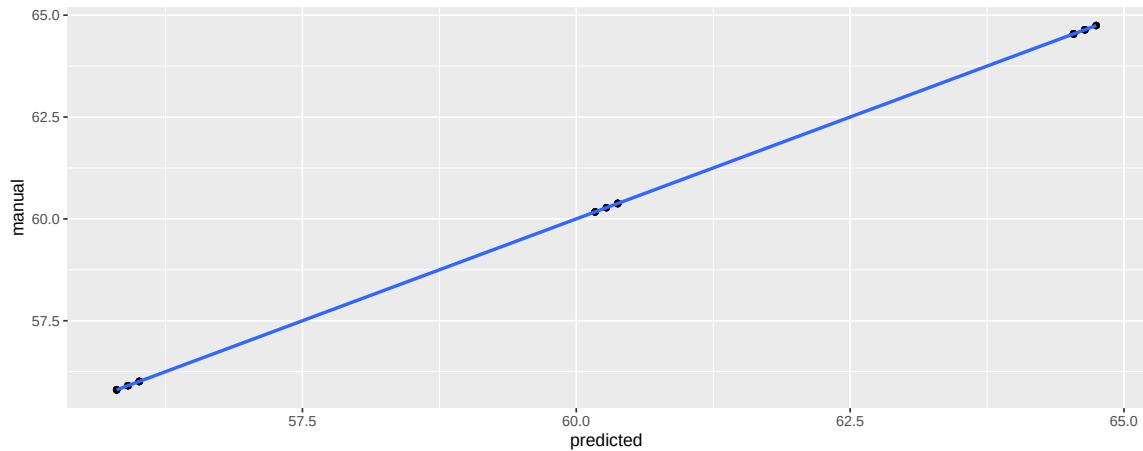
```
( paras <- coef( model2 ) );
```

(Intercept)	class	age
52.880418	4.471066	-0.103170

```
predicted2 <-
  paras[1] + paras["age"] * wts2$age + paras["class"] * wts2$class;
plot <-
  ggplot( data.frame( predicted = wt.predicted, manual = predicted2 ),
    aes( predicted, manual ) ) +
    geom_point() + geom_smooth( method = "lm", se = F );
```

show the plot

```
plot;
```



更多更方便的函数

以下函数也可以用于 multivariable analysis / multiple regression

```
fit <- lm(y ~ x1 + x2 + x3, data=mydata)
summary(fit) # show results

# Other useful functions
coefficients(fit) # model coefficients
confint(fit, level=0.95) # CIs for model parameters
fitted(fit) # predicted values
residuals(fit) # residuals
anova(fit) # anova table
vcov(fit) # covariance matrix for model parameters
influence(fit) # regression diagnostics
```

linear regression 注意事项

- ① 是 parametric test
- ② 假设变量之间独立（比如：年龄和班级之间没有关联）
- ③ homogeneity of variance

但实际上 ...

multivariable analysis

```
# Multiple Linear Regression Example
fit <- lm(y ~ x1 + x2 + x3, data=mydata)
summary(fit) # show results

# compare models
fit1 <- lm(y ~ x1 + x2 + x3 + x4, data=mydata)
fit2 <- lm(y ~ x1 + x2)
anova(fit1, fit2)

# K-fold cross-validation
library(DAAG)
cv.lm(df=mydata, fit, m=3) # 3 fold cross-validation

# Stepwise Regression; feature selection
library(MASS)
fit <- lm(y~x1+x2+x3,data=mydata)
step <- stepAIC(fit, direction="both")
step$anova # display results
```

extended reading

- ① repeated measures ANOVA , 同一变量、不同时间段的重复测量 (对上例中学生的体重进行多次测量)
- ② correlation and linear regression
- ③ non-linear regression

Section 6

section 6: non-parametric test

wilcox.test and kruskal.test

```
# independent 2-group Mann-Whitney U Test  
with( Data, wilcox.test( Steps ~ Sex ) );
```

Wilcoxon rank sum test with continuity correction

data: Steps by Sex

W = 127.5, p-value = 0.01773

alternative hypothesis: true location shift is not equal to 0

```
# Kruskal Wallis Test One Way Anova by Ranks  
with( Data, kruskal.test( Steps ~ Sex ) );
```

Kruskal-Wallis rank sum test

data: Steps by Sex

Kruskal-Wallis chi-squared = 5.7494, df = 1, p-value = 0.01649

Section 7

作业与练习

作业与练习

- Exercises and homework 目录下 talk10-homework.Rmd 文件
- 完成时间: 见钉群的要求